

**Notes: Pankratz & Schiller (RFS, 2024)**  
**Climate Change and Adaptation in Global Supply-Chain Networks**

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# 1 Big picture

- **Question.** How does physical climate exposure at supplier locations affect firm performance, supply-chain stability, and the formation of new supplier relationships?
- **Main setting.** The paper studies global customer–supplier links from FactSet Revere over 2003–2016, matched to firm financials, supplier/customer locations, local temperatures, climate projections, and adaptation measures.
- **Direct financial effect.** Heat at supplier locations reduces suppliers’ operating income over assets. The most conservative specification implies that one additional hot day lowers supplier quarterly operating income/assets by roughly 0.0087 percentage points summed over current and lagged quarters.
- **Propagation effect.** Heat at suppliers also lowers the operating income of downstream customers. The effect is smaller than the direct supplier effect but economically meaningful for customers because exposure is transmitted through production networks.
- **Core adaptation result.** Customers are more likely to terminate a supplier relationship when realized heat exposure exceeds the supplier’s historical expected exposure.
- **Learning interpretation.** The effect is stronger when the signal is stronger or repeated and weaker in supplier countries with greater climate-change adaptation readiness.
- **Not merely a one-off disruption.** Contemporaneous realized heat days alone do not predict termination once expectations are considered, which supports the interpretation that customers respond to perceived distributional changes rather than transitory shocks only.
- **Other adjustment margins.** Customers increase inventories, cost of goods sold, R&D, and cash after perceived supplier exposure rises, and they increase supplier diversification in related industries.
- **Replacement suppliers.** After terminating exposed suppliers, customers choose replacement suppliers with lower expected and realized heat exposure during and after the initial relationship. Effects are weaker for long-run projected heat exposure.
- **Economic takeaway.** Climate change affects firm boundaries and supply-chain organization: firms learn from experienced climate realizations and adapt by reshaping supplier networks.

## 2 Data and measurement

### 2.1 Supply-chain relationships

- Supplier-customer links come from FactSet Revere and cover global listed firms from 2003 to 2016.
- Financial data are from Worldscope. Headquarters and establishment locations come from FactSet Fundamentals and Orbis.
- The unit of analysis varies by test: supplier-year-quarter, customer-year-quarter, supplier-customer pair-year, or replacement supplier comparison.
- The baseline excludes financial firms and firms with less than 10% of locations within 30 km of headquarters to reduce measurement error in assigning exposure.
- Customers located within 500 km of affected suppliers are excluded in key customer tests, so results are less likely to reflect common local demand or infrastructure shocks.

**Interpretation.** The data combine network links, financial outcomes, and geocoded climate exposure. This lets the paper measure both direct heat effects on suppliers and indirect effects on their customers.

## 2.2 Heat exposure

- A heat day is a day with local temperature above 30°C at the supplier location.
- For financial-performance tests, heat exposure is measured quarterly and includes current and lagged quarters.
- For supply-chain adaptation tests, exposure is measured annually and compared with historical expectations before a supplier relationship begins.
- The authors also study floods in the Internet Appendix and find qualitatively similar patterns.

**Interpretation.** Heat is a noisy but observable signal. The paper distinguishes realized heat shocks from perceived changes in exposure relative to what customers could have expected ex ante.

## 2.3 Expected versus realized exposure

Let  $ExpectedExposure_{sc}$  be the historical average number of heat days per year at supplier  $s$  before the start of relationship  $(s, c)$ , usually estimated over a ten-year benchmark window. Let  $RealizedExposure_{sct}$  be the average realized number of heat days per year since the start of the relationship.

$$Exceed_{sct} = \mathbf{1}\{RealizedExposure_{sct} > ExpectedExposure_{sc}\}.$$

**Interpretation.** The treatment is not raw heat. It is a deviation from a relationship-specific historical expectation, which captures a perceived increase in supplier climate exposure.

## 2.4 Outcomes

- **Supplier/customer performance:** operating income scaled by lagged assets, multiplied by 100.
- **Relationship termination:** indicator for the last year of an active supplier-customer link.
- **Other adjustments:** inventory, COGS, R&D, cash, supplier diversification, and acquisitions.
- **Substitution:** indicators that a replacement supplier has lower heat exposure than the terminated supplier during, after, or in projected future periods.

**Interpretation.** The empirical design links exposure first to financial incentives and then to concrete supply-chain adaptation choices.

# 3 Models and empirical specifications

## 3.1 Direct effect of heat on suppliers

The supplier financial-performance regression is

$$y_{it} = \sum_{\ell=0}^3 \beta_{\ell} W_{i,t-\ell} + \mu_{iq} + \gamma_{n(i)t} + \theta_{d(i)t} + \delta' BS2016_{it} + \varepsilon_{it},$$

where  $y_{it}$  is supplier operating income/assets,  $W_{i,t-\ell}$  is the number of heat days at the supplier,  $\mu_{iq}$  are supplier-by-fiscal-quarter fixed effects,  $\gamma_{n(i)t}$  are industry-by-year-quarter fixed effects, and  $\theta_{d(i)t}$  are country-specific linear trends.

**Interpretation.** This specification uses within-firm seasonal variation in heat while absorbing industry-time, country-trend, and firm-profile trends.

### 3.2 Propagation from suppliers to customers

The customer performance regression is

$$y_{ct} = \sum_{\ell=0}^3 \beta_{\ell} W_{c,t-\ell} + \mu_{cq} + \gamma_{n(c)t} + \theta_{d(c)t} + \delta' BS2016_{ct} + \varepsilon_{ct},$$

where  $W_{c,t-\ell}$  aggregates heat exposure across customer  $c$ 's suppliers.

**Interpretation.** A negative coefficient means supplier-location heat reduces downstream customer performance, providing a financial incentive for customer adaptation.

### 3.3 Termination response to exceeded expectations

The main relationship-level specification is

$$\mathbf{1}\{End_{sct}\} = \beta Exceed_{sct} + \gamma_{n(s)t} + \gamma_{n(c)t} + \theta_{d(s)d(c)t} + \varepsilon_{sct}.$$

- $\gamma_{n(s)t}$  and  $\gamma_{n(c)t}$  absorb supplier- and customer-industry-year trends.
- $\theta_{d(s)d(c)t}$  absorbs supplier-country by customer-country by year shocks.
- The coefficient  $\beta$  measures how much more likely a relationship is to end after supplier exposure exceeds historical expectations.

**Interpretation.** The design compares relationships exposed to unexpectedly high heat with other relationships in the same country-pair-year and industry-year environments.

### 3.4 Signal strength, repetition, and nonlinearity

The paper estimates versions of the main regression using indicators for repeated years of exceeded expectations and continuous deviations:

$$\mathbf{1}\{End_{sct}\} = f(RealizedExposure_{sct} - ExpectedExposure_{sc}) + FE + \varepsilon_{sct}.$$

**Interpretation.** If customers learn from experience, one weak signal should matter less than repeated or stronger deviations. Table 3 shows exactly this pattern.

### 3.5 Replacement supplier specification

For replacement suppliers, the paper estimates

$$\mathbf{1}\{ExposureNew < ExposureOld\}_{sc} = \beta Exceed_{st} + \gamma_{n(s)t} + \gamma_{n(c)t} + \theta_{d(s)t} + \theta_{d(c)t} + \varepsilon_{sc}.$$

**Interpretation.** This tests whether firms not only terminate exposed relationships but also actively choose less exposed replacement suppliers.

## 4 Identification and empirical design

### 4.1 Why the heat-performance tests are credible

- Within-location quarterly heat variation is plausibly exogenous conditional on fixed effects.
- Supplier-by-fiscal-quarter fixed effects remove firm-specific seasonality.
- Industry-by-time fixed effects remove sector-level cycles.
- Country trends remove simultaneous development and climate trends.
- Customer tests exclude geographically proximate supplier-customer pairs to reduce common local-shock concerns.

**Interpretation.** The heat-performance results establish that suppliers' physical exposure creates real financial consequences for both suppliers and customers.

### 4.2 Why expected-versus-realized exposure matters

- Long-run climate trends are not randomly assigned, but short-run deviations from local historical expectations are noisy and hard to forecast.
- Managers may incorporate historical expected exposure into the original supplier choice.
- If realized exposure exceeds that benchmark during the relationship, customers receive a signal that supplier exposure may have changed.

**Interpretation.** The treatment captures learning about climate risk rather than simply operating in a hot place.

### 4.3 Fixed effects and exclusion restrictions

- Supplier- and customer-industry-year fixed effects absorb time-varying sector shocks.
- Supplier-country by customer-country by year fixed effects absorb trade barriers, macro shocks, and bilateral conditions.
- The design excludes local customer-supplier pairs, so effects are less likely due to the customer's own local heat exposure.
- The paper does not include relationship fixed effects in the main termination regression because the treatment captures within-relationship updating and the outcome is relationship ending.

**Interpretation.** For an omitted variable to explain the result, it would need to vary within industry-year and country-pair-year in exactly the relationships where realized supplier exposure exceeds expectations.

### 4.4 Mechanism and falsification checks

- Table 4 shows that transitory heat days alone do not meaningfully predict relationship termination.
- Table 5 shows customers also adjust inventories, R&D, cash, and supplier diversification.
- Table 7 shows the termination response is weaker in countries with stronger climate adaptation readiness.
- Table 9 and Figure 5 show customers substitute toward suppliers with lower expected and realized heat exposure.

**Interpretation.** The evidence is consistent with experience-based learning and operational adaptation, not just one-off production disruptions.

## 4.5 Limitations

- The authors do not observe managers’ subjective beliefs directly.
- Supply-chain links are observed for large listed firms and disclosed relationships; private or undisclosed suppliers may be missing.
- Termination can reflect several mechanisms, including learning, disruptions, renegotiation, or strategic sourcing.
- Climate projections are long-horizon and coarse relative to managers’ immediate sourcing decisions.

**Interpretation.** The strongest evidence is the full pattern: financial damages, termination after exceeded expectations, heterogeneity by learning/adaptation channels, and substitution toward less exposed suppliers.

## 5 Tables and figures

### 5.1 Figure 1 and Table 1: sample geography and descriptive statistics

**Read:**

- Figure 1 shows a dense global network of customer-supplier links, with suppliers concentrated in Asia and customers spread across North America, Europe, and Asia.
- Table 1 reports that the average supplier experiences about 14 heat days per quarter and has an average annual temperature of about 19°C.
- The median customer is much larger than the median supplier: customers hold roughly 12 times the assets of the median supplier.
- Supplier-customer links are economically important but often not dominant: suppliers account for about 2% of customers’ COGS on average in disclosed relationships.

**Economic message.** The paper studies geographically dispersed production networks where supplier climate exposure can transmit shocks to globally connected customers.

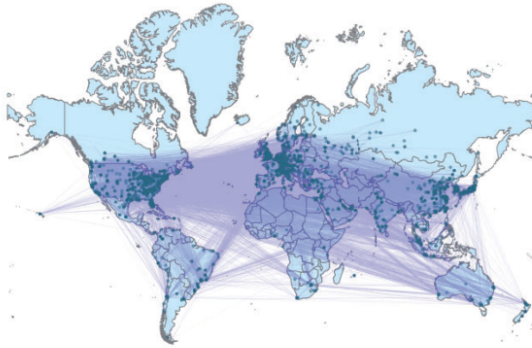


Figure 1

Figure 1: global customer-supplier links

Table 1  
(Continued)

| (f) Unique supplier-year-quarter observations |        |        |        |       |       |        |
|-----------------------------------------------|--------|--------|--------|-------|-------|--------|
|                                               | N      | Mean   | StDev  | p25   | p50   | p75    |
| Supplier Heat Days (t)                        | 202439 | 14.091 | 23.624 | 0.000 | 1.000 | 18.000 |
| Heat Days ex. EM-DAT Heatwave                 | 202439 | 13.559 | 23.215 | 0.000 | 0.000 | 17.000 |
| Heat Days ex. EM-DAT Fire                     | 202439 | 13.296 | 23.267 | 0.000 | 0.000 | 16.000 |
| Heat Days ex. EM-DAT Drought                  | 202439 | 13.960 | 23.584 | 0.000 | 1.000 | 18.000 |
| Heat Days ex. EM-DAT Heatwave/Drought/Fire    | 202439 | 12.635 | 22.792 | 0.000 | 0.000 | 14.000 |

This table presents summary statistics at the quarterly level for suppliers (a,e) and customers (b) as well as at the yearly level for customer-supplier pairs (c), suppliers (d), and customers (e) in our sample. The sample period is 2003 to 2016. Table A.1 in the appendix defines all variables. Firm-level accounting data are from Worldscope. “Asset tangibility” is the ratio of PPE to total assets. “Ind. vulnerability” is an indicator that takes the value of one if the firm operates in a high climate-risk industry. “Industry Competitiveness” is the number of firms (in thousands) per SIC two-digit industry. “BEA Input-Ind. Concentration” is the Herfindahl-Hirschman index (HHI) of inputs per industry from the BEA Input-Output matrices. “Supplier diversification” is the ratio of the

Table 1f: heat days and disaster overlap

**Table 1**  
Summary statistics

| (a) Unique supplier-year-quarter observations |         |          |          |          |          |          |
|-----------------------------------------------|---------|----------|----------|----------|----------|----------|
|                                               | N       | Mean     | SD       | p25      | Median   | p75      |
| Total assets (US\$ billions)                  | 202,439 | 3.145    | 13.214   | 0.114    | 0.399    | 1.457    |
| Revenue/Assets (%)                            | 202,439 | 26.660   | 20.664   | 12.224   | 21.859   | 35.221   |
| Operating income/Assets (%)                   | 202,439 | 1.020    | 4.664    | -0.037   | 1.414    | 3.086    |
| Heat days (30° C)                             | 202,439 | 14.091   | 23.624   | 0.000    | 1.000    | 18.000   |
| Heat days (within-location)                   | 202,439 | -0.000   | 16.170   | -8.967   | -1.128   | 3.571    |
| Heat days (30° C-95P)                         | 202,439 | 3.146    | 5.788    | 0.000    | 0.000    | 4.000    |
| Heatwave (30° C/7 days)                       | 202,439 | 0.243    | 0.429    | 0.000    | 0.000    | 0.000    |
| Heatwave (30° C-95P/7 days)                   | 202,439 | 0.032    | 0.175    | 0.000    | 0.000    | 0.000    |
| Average temperature                           | 202,439 | 19.292   | 5.591    | 15.016   | 19.197   | 22.325   |
| Heat days (EM-DAT)                            | 202,439 | 0.663    | 4.129    | 0.000    | 0.000    | 0.000    |
| (b) Unique customer-year-quarter observations |         |          |          |          |          |          |
|                                               | N       | Mean     | SD       | p25      | Median   | p75      |
| Total assets (US\$ billions)                  | 124,140 | 18.533   | 45.177   | 0.883    | 3.284    | 12.279   |
| Market capitalization (US\$ billions)         | 123,379 | 9.166    | 16.117   | 0.675    | 2.608    | 9.140    |
| Revenue/Assets (%)                            | 125,359 | 25.757   | 20.339   | 11.788   | 21.137   | 33.914   |
| Operating income/Assets (%)                   | 125,359 | 1.802    | 3.132    | 0.472    | 1.665    | 3.203    |
| (c) Unique firm-pair-year observations        |         |          |          |          |          |          |
|                                               | N       | Mean     | StDev    | p25      | p50      | p75      |
| I (Last Relationship Year)                    | 131817  | 0.270    | 0.444    | 0.000    | 0.000    | 1.000    |
| Relationship Age (Years)                      | 131207  | 5.079    | 3.870    | 2.000    | 4.000    | 7.000    |
| Pct. Sales Sup (%)                            | 13924   | 19.382   | 18.141   | 10.000   | 14.000   | 23.000   |
| Pct. COGS Cus (%)                             | 11852   | 2.358    | 7.370    | 0.072    | 0.297    | 1.288    |
| Sup-Cus HQ Distance                           | 130964  | 5544.352 | 4091.436 | 1707.374 | 4305.921 | 8815.658 |
| MCap Cus / MCap Sup                           | 109036  | 223.363  | 457.354  | 4.665    | 29.512   | 165.017  |
| Assets Cus / Assets Sup                       | 110817  | 312.119  | 614.817  | 7.074    | 43.905   | 249.268  |
| Sales Correlation                             | 81465   | 0.149    | 0.425    | -0.160   | 0.170    | 0.476    |
| 1(Realized > Exp.) HeatDays (> 1)             | 131817  | 0.399    | 0.490    | 0.000    | 0.000    | 1.000    |
| 1(Realized > Exp.) HeatDays                   | 131817  | 0.513    | 0.500    | 0.000    | 1.000    | 1.000    |
| RCP2.6 & Heat Days 2006/19-2040/59            | 131811  | 7.026    | 8.385    | 2.186    | 6.202    | 9.833    |
| RCP4.5 & Heat Days 2006/19-2040/59            | 131811  | 11.781   | 13.375   | 3.071    | 9.462    | 15.124   |
| RCP8.5 & Heat Days 2006/19-2040/59            | 131811  | 22.046   | 18.131   | 9.431    | 21.507   | 27.133   |
| (d) Unique supplier-year observations         |         |          |          |          |          |          |
|                                               | N       | Mean     | StDev    | p25      | p50      | p75      |
| Asset Tangibility                             | 22457   | 0.259    | 0.243    | 0.062    | 0.175    | 0.396    |
| Ind. Vulnerability                            | 23951   | 0.067    | 0.249    | 0.000    | 0.000    | 0.000    |
| No. of Customers                              | 23927   | 6.901    | 8.015    | 2.000    | 4.000    | 9.000    |
| Ind. Competitiveness                          | 23949   | 0.832    | 0.656    | 0.323    | 0.601    | 1.318    |
| Ind. HHI MCap                                 | 23949   | 0.106    | 0.108    | 0.027    | 0.060    | 0.192    |
| (e) Unique customer-year observations         |         |          |          |          |          |          |
|                                               | N       | Mean     | StDev    | p25      | p50      | p75      |
| BEA Input-Ind. Concentration                  | 14262   | 0.051    | 0.107    | 0.011    | 0.022    | 0.040    |
| Supplier Diversification                      | 26576   | 1.783    | 1.272    | 1.000    | 1.286    | 2.000    |
| No. Suppliers                                 | 29278   | 9.024    | 19.847   | 1.000    | 3.000    | 8.000    |
| No. Suppliers / Assets                        | 26894   | 5.603    | 26.951   | 0.503    | 1.189    | 3.009    |
| Acct. Payable / Assets                        | 23223   | 0.109    | 0.101    | 0.039    | 0.078    | 0.143    |
| Acct. Receivable / Assets                     | 23340   | 0.164    | 0.127    | 0.069    | 0.134    | 0.224    |
| COGS / Assets                                 | 23253   | 0.669    | 0.561    | 0.241    | 0.526    | 0.913    |
| Inventory / Assets                            | 23286   | 0.116    | 0.114    | 0.021    | 0.088    | 0.174    |

(Continued)

## 5.2 Table 2: physical heat exposure and financial incentives to adapt

Read:

- Panel A shows that heat at supplier locations lowers supplier operating income/assets.
- In the most conservative supplier specification, the combined current and lagged effect is about -0.0087 percentage points per extra heat day.
- A one-standard-deviation increase in heat exposure translates into roughly a 13.8% decrease in supplier operating income/assets.
- Panel B shows downstream propagation: supplier heat lowers customer operating income/assets by about -0.0007 percentage points summed over current and lagged quarters.
- The customer effect is smaller, about 4.6% of the direct supplier effect, but it is economically meaningful for large customers.

**Economic message.** Climate exposure at supplier sites creates financial incentives for customers to monitor and adapt their supplier networks.

**Table 2**  
Physical exposure to heat and financial incentives to adapt

| (a) Direct Effect on Supplier |                       |                       |                      |                      | (b) Propagation to Customer |                        |                        |                        |                        |
|-------------------------------|-----------------------|-----------------------|----------------------|----------------------|-----------------------------|------------------------|------------------------|------------------------|------------------------|
|                               | Sup Opl (t)           |                       |                      |                      |                             | Cus Opl (t)            |                        |                        |                        |
|                               | (1)                   | (2)                   | (3)                  | (4)                  |                             | (1)                    | (2)                    | (3)                    | (4)                    |
| Supplier Heat Days (t)        | -0.0039***<br>(-2.96) | -0.0047***<br>(-3.48) | -0.0032**<br>(-2.38) | -0.0033**<br>(-2.40) | Supplier Heat Days (t)      | -0.0003***<br>(-4.422) | -0.0002***<br>(-2.921) | -0.0002***<br>(-2.683) | -0.0001*<br>(-1.933)   |
| Supplier Heat Days (t-1)      | -0.0018<br>(-1.39)    | -0.0035***<br>(-2.64) | -0.0025*<br>(-1.83)  | -0.0027*<br>(-1.95)  | Supplier Heat Days (t-1)    | -0.0006***<br>(-6.553) | -0.0004***<br>(-4.225) | -0.0003***<br>(-4.010) | -0.0003***<br>(-3.196) |
| Supplier Heat Days (t-2)      | -0.0039***<br>(-3.01) | -0.0044***<br>(-3.22) | -0.0031**<br>(-2.32) | -0.0027**<br>(-2.02) | Supplier Heat Days (t-2)    | -0.0003***<br>(-4.145) | -0.0001**<br>(-2.129)  | -0.0001*<br>(-1.851)   | -0.0001**<br>(-2.025)  |
| Supplier Heat Days (t-3)      | -0.0024*<br>(-1.77)   | -0.0023*<br>(-1.67)   | -0.0015<br>(-1.05)   | -0.0015<br>(-1.05)   | Supplier Heat Days (t-3)    | -0.0005***<br>(-7.214) | -0.0003***<br>(-3.760) | -0.0002***<br>(-3.380) | -0.0002***<br>(-3.092) |
| Firm × Fiscal-Qtr FE          | Yes                   | Yes                   | Yes                  | Yes                  | Firm × Fiscal-Qtr FE        | Yes                    | Yes                    | Yes                    | Yes                    |
| Ind × Year-Qtr FE             | No                    | Yes                   | Yes                  | Yes                  | Ind × Year-Qtr FE           | No                     | Yes                    | Yes                    | Yes                    |
| Ctry-Linear-Trends            | No                    | No                    | Yes                  | Yes                  | Ctry-Linear-Trends          | No                     | No                     | Yes                    | Yes                    |
| BS2016 FE                     | No                    | No                    | No                   | Yes                  | BS2016 FE                   | No                     | No                     | No                     | Yes                    |
| Observations                  | 202438                | 202438                | 202438               | 202438               | Observations                | 123700                 | 123700                 | 123700                 | 123700                 |
| Customers                     | 5628                  | 5628                  | 5628                 | 5628                 | Customers                   | 6299                   | 6299                   | 6299                   | 6299                   |
| R <sup>2</sup>                | .612                  | .623                  | .626                 | .631                 | R <sup>2</sup>              | .69                    | .706                   | .707                   | .711                   |

This table presents OLS regression estimates on the effects of heat at the location of the sample supplier firms on the operating income (Opl), scaled by lagged assets, of suppliers (panel 2a) and their downstream customers (panel 2b). The dependent variable is multiplied by 100 for ease of interpretation. *Heat days (t)* indicate the number of hot days at the supplier firm during the financial quarter *t* and the three preceding quarters (*t* − 3 to *t* − 1). The number of observations refers to supplier firm year-quarters (panel 2a) and customer firm year-quarters (panel 2b). In panel 2b, we aggregate the number of supplier heat days for each customer year-quarter. The sample period is 2003 to 2016. We exclude firms in the financial industry as well as firms with less than 10% of firm locations within 30 kilometers of the company headquarter. The regressions include firm-by-fiscal quarter fixed effects to control for time invariant firm characteristics and firm-specific seasonal effects, industry-by-year-by-quarter fixed effects, controls for country-specific linear trends, and interaction terms of terciles of firm size, age, and ROA with year-by-quarter fixed effects to control for firm characteristics following Barrot and Sauvagnat (2016) as indicated. Standard errors are clustered at the firm level. \*  $p < .1$ ; \*\*  $p < .05$ ; \*\*\*  $p < .01$ .

### 5.3 Figures 2 and 3: variable construction and identifying variation

Read:

- Figure 2 illustrates the main treatment: historical expected heat is estimated before the relationship, and realized exposure is averaged during the relationship.
- The exceedance indicator turns on only when realized annual heat days rise above the historical benchmark.
- Figure 3 shows that the distribution of realized-minus-expected heat is similar before and after absorbing high-dimensional fixed effects.
- This supports the view that deviations from historical expectations are not driven by simple country-year or industry-year trends.

**Economic message.** The treatment variable is designed to measure perceived increases in exposure, not the mere level of heat in hot countries.

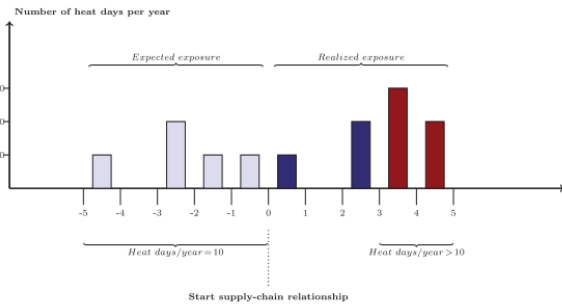


Figure 2

Figure 2: expected versus realized exposure

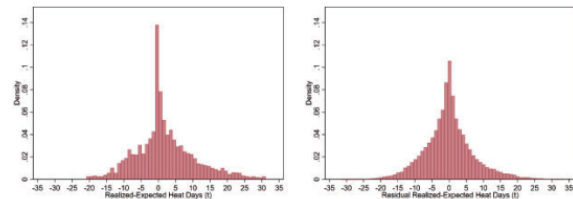


Figure 3

Figure 3: identifying variation



## 5.4 Table 3 and Table 4: termination and transitory heat

**Read:**

- Table 3 shows that relationships are more likely to end when realized heat exposure exceeds expectations.
- The main linear probability estimate is about 1.013 percentage points, while the logit estimate implies about a 7.4% increase in termination likelihood.
- The effect becomes especially strong after repeated exceeded expectations: the second and third years of exceedance have large positive effects.
- Table 4 shows that contemporaneous realized heat days alone are economically small and statistically insignificant across specifications.

**Economic message.** Customers do not appear to terminate suppliers merely because a hot year occurs. They respond when realized exposure changes relative to what they could have expected.

**Table 3**  
Expected versus realized exposure to heat and relationship termination

| (a) Robust SE: Relationship clusters  |                     |                      |                     |
|---------------------------------------|---------------------|----------------------|---------------------|
|                                       | I (Last Rel. Year)  |                      |                     |
|                                       | (1)<br>OLS          | (2)<br>Logit         | (3)<br>OLS          |
| I (Real.>Exp. Heat Days)              | 1.013***<br>(3.982) |                      |                     |
| I (Real.>Exp. Heat Days) (GLM)        |                     | 0.0741***<br>(4.113) |                     |
| I (Real.>Exp. Heat Days) (> 1)        |                     |                      | 7.520***<br>(26.72) |
| Sup.-Ind.-Year FE                     | Yes                 | Yes                  | Yes                 |
| Cus.-Ind.-Year FE                     | Yes                 | Yes                  | Yes                 |
| Sup-Ctry-Cus-Ctry-Year FE             | Yes                 | Yes                  | Yes                 |
| Observations                          | 116,412             | 104,581              | 116,412             |
| R <sup>2</sup>                        | 0.1290              |                      | 0.1356              |
| (b) Robust SE: Rel. and year clusters |                     |                      |                     |
|                                       | I (Last Rel. Year)  |                      |                     |
|                                       | (1)<br>OLS          | (2)<br>Logit         | (3)<br>OLS          |
| I (Real.>Exp. Heat Days)              | 1.013<br>(0.9493)   |                      |                     |
| I (Real.>Exp. Heat Days) (GLM)        |                     | 0.0741<br>(0.9479)   |                     |
| I (Real.>Exp. Heat Days) (> 1)        |                     |                      | 7.520***<br>(6.237) |
| Sup.-Ind.-Year FE                     | Yes                 | Yes                  | Yes                 |
| Cus.-Ind.-Year FE                     | Yes                 | Yes                  | Yes                 |
| Sup-Ctry-Cus-Ctry-Year FE             | Yes                 | Yes                  | Yes                 |
| Observations                          | 116,412             | 104,581              | 116,412             |

Table 3a-b: termination after exceeded expectations

**Table 4**  
Relationship termination in response to transitory heat exposure

|                           | OLS - I (Last Relationship Year) |                   |                    |                    |                    |
|---------------------------|----------------------------------|-------------------|--------------------|--------------------|--------------------|
|                           | (1)                              | (2)               | (3)                | (4)                | (5)                |
| Realized Heat Days        | 0.0333<br>(1.561)                | 0.0235<br>(1.033) | 0.0222<br>(0.9207) | 0.0114<br>(0.4327) | 0.0114<br>(0.4327) |
| Supplier Firm FE          | Yes                              | Yes               | Yes                | Yes                | Yes                |
| Year FE                   | Yes                              | Yes               | Yes                | Yes                | Yes                |
| Sup.-Ctry-Year FE         | No                               | Yes               | Yes                | Yes                | Yes                |
| Sup-Ctry-Cus-Ctry-Year FE | No                               | No                | Yes                | Yes                | Yes                |
| Sup.-Ind.-Year FE         | No                               | No                | No                 | Yes                | Yes                |
| Cus.-Ind.-Year FE         | No                               | No                | No                 | Yes                | Yes                |
| Observations              | 116,412                          | 116,412           | 116,412            | 116,412            | 116,412            |
| R <sup>2</sup>            | 0.1639                           | 0.2158            | 0.2618             | 0.2666             | 0.2666             |

This table presents linear probability model estimates on the effect of *transitory* heat occurrence on the likelihood of relationship termination. *Realized Heat Days* measures the number of hot days at the supplier locations in

Table 4: transitory heat exposure

**Table 3**  
(Continued)

|                                          | (c) Magnitude of deviation |                   |                      |
|------------------------------------------|----------------------------|-------------------|----------------------|
|                                          | 1(Last Rel. Year)          |                   |                      |
|                                          | (1)                        | (2)               | (3)                  |
| 1(Real.>Exp. Heat Days) = 1              | -1.432<br>(-1.122)         |                   |                      |
| 1(Real.>Exp. Heat Days) = 2              | 9.342***<br>(5.164)        |                   |                      |
| 1(Real.>Exp. Heat Days) = 3              | 6.426***<br>(3.085)        |                   |                      |
| 1(Real.>Exp. Heat Days) = 4              | 4.951*<br>(2.037)          |                   |                      |
| 1(Real.>Exp. Heat Days) = 5              | 0.9254<br>(0.7566)         |                   |                      |
| (Real.>Exp. Heat Days) (>1)              |                            | 0.1239<br>(1.719) | -0.0651<br>(-1.308)  |
| (Real.>Exp. Heat Days) <sup>2</sup> (>1) |                            |                   | 0.0269***<br>(7.171) |
| Sup.-Ctry.-Year FE                       | Yes                        | Yes               | Yes                  |
| Sup.-Ind.-Year FE                        | Yes                        | Yes               | Yes                  |
| Cus.-Ind.-Year FE                        | Yes                        | Yes               | Yes                  |
| Sup-Ctry-Cus-Ctry.-Year FE               | Yes                        | Yes               | Yes                  |
| Observations                             | 116,412                    | 115,298           | 115,298              |
| R <sup>2</sup>                           | 0.1364                     | 0.1290            | 0.1322               |

This table presents estimates of the effect of 1(*Realized > Expected*)(*t*) on the likelihood of supply-chain

## 5.5 Tables 5 and 6: other adjustments and cross-sectional mechanisms

### Read:

- Table 5 shows that customers increase inventories, COGS, R&D, and cash after suppliers' heat exposure exceeds expectations.
- Customers also increase the number of suppliers in the same two-digit supplier industry, suggesting diversification rather than pure replacement only.
- Table 6 shows stronger termination effects when relationships are less integrated, shorter, and less protected by supplier diversification.
- The response is stronger when supplier industries have more firms and lower sales concentration, consistent with easier switching when alternatives are available.

**Economic message.** Firms adapt through multiple operational margins, and the termination response is strongest when switching costs are lower and substitutes are easier to find.

Table 5

## Other adjustments and supplier diversification

| (a) Other margins of adjustment |                     |                     |                      |                      |
|---------------------------------|---------------------|---------------------|----------------------|----------------------|
|                                 | Inventory<br>(1)    | COGS<br>(2)         | R&D<br>(3)           | Cash<br>(4)          |
| I(Real.>Exp. Heat Days)         | 0.0071*<br>(2.095)  | 0.0092**<br>(2.451) | 0.0198***<br>(4.434) | 0.0246***<br>(3.973) |
| Customer Firm FE                | Yes                 | Yes                 | Yes                  | Yes                  |
| Cus.-Ind.-Year FE               | Yes                 | Yes                 | Yes                  | Yes                  |
| Cus.-Ctry-Year FE               | Yes                 | Yes                 | Yes                  | Yes                  |
| Observations                    | 21,912              | 23,855              | 13,183               | 21,325               |
| R <sup>2</sup>                  | 0.9774              | 0.9765              | 0.9755               | 0.8727               |
| (b) Supplier diversification    |                     |                     |                      |                      |
|                                 | Sup. SIC2<br>(1)    | Sup. SIC3<br>(2)    | Acq. SIC2<br>(3)     | Acq. SIC3<br>(4)     |
| I(Real.>Exp. Heat Days)         | 0.0552**<br>(2.334) | 0.0296<br>(1.521)   | -0.0043<br>(-1.331)  | -0.0026<br>(-0.9836) |
| Sup.-Ind.-Year FE               | Yes                 | Yes                 | Yes                  | Yes                  |
| Cus.-Ind.-Year FE               | Yes                 | Yes                 | Yes                  | Yes                  |
| Sup-Ctry-Cus-Ctry-Year FE       | Yes                 | Yes                 | Yes                  | Yes                  |
| Observations                    | 76,057              | 76,062              | 111,918              | 111,918              |
| R <sup>2</sup>                  | 0.2105              | 0.1757              | 0.1095               | 0.1036               |

This table presents OLS estimates on the effect of supplier heat exposure on customer firm risk management and

Table 5: other adjustment margins

Table 6

## Supply-chain integration, industry reliance, and supplier competition

|                                                            | Dep. Var: 1(Last Relationship Year) |                       |                        |                     |                     |                     |                        |
|------------------------------------------------------------|-------------------------------------|-----------------------|------------------------|---------------------|---------------------|---------------------|------------------------|
|                                                            | Cus.-Sup. Integration               |                       | Ind. Reliance          |                     | Sup. Competition    |                     |                        |
|                                                            | (1)                                 | (2)                   | (3)                    | (4)                 | (5)                 | (6)                 | (7)                    |
| I(Real.>Exp. Heat Days) (> 1)                              | 16.47***<br>(3.525)                 | 15.02***<br>(7.288)   | 10.35***<br>(6.980)    | 13.17**<br>(2.874)  | 9.511***<br>(4.359) | 5.432***<br>(3.420) | 7.525***<br>(6.241)    |
| I(Real.>Exp. Heat Days) (> 1)<br>× Log(Sales to Cus)       | -0.8276**<br>(-2.932)               |                       |                        |                     |                     |                     |                        |
| I(Real.>Exp. Heat Days) (> 1)<br>× Rel. Length             |                                     | -2.838***<br>(-8.618) |                        |                     |                     |                     |                        |
| I(Real.>Exp. Heat Days) (> 1)<br>× Sup. Diversification    |                                     |                       | -0.8350***<br>(-5.884) |                     |                     |                     |                        |
| I(Real.>Exp. Heat Days) (> 1)<br>× Sup.-to-Cus.-Ind. Sales |                                     |                       |                        | -15.14*<br>(-1.843) |                     |                     |                        |
| I(Real.>Exp. Heat Days) (> 1)<br>× HHI Sup.-Ind. Inputs    |                                     |                       |                        |                     | -6.619*<br>(-2.113) |                     |                        |
| I(Real.>Exp. Heat Days) (> 1)<br>× N Firms Sup.-Ind.       |                                     |                       |                        |                     |                     | 1.594**<br>(2.281)  |                        |
| I(Real.>Exp. Heat Days) (> 1)<br>× HHI Sales Sup.-Ind.     |                                     |                       |                        |                     |                     |                     | -0.0861***<br>(-6.145) |
| Log(Sales to Cus)                                          | -1.592***<br>(-5.373)               |                       |                        |                     |                     |                     |                        |
| Rel. Length                                                |                                     | 1.487***<br>(8.915)   |                        |                     |                     |                     |                        |
| Sup. Diversification                                       |                                     |                       | -0.5014***<br>(-6.077) |                     |                     |                     |                        |
| Sup.-to-Cus.-Ind. Sales                                    |                                     |                       |                        | -1.651<br>(-0.2919) |                     |                     |                        |
| HHI Sup.-Ind. Inputs                                       |                                     |                       |                        |                     | -2.229<br>(-0.9693) |                     |                        |
| N Firms Sup.-Ind.                                          |                                     |                       |                        |                     |                     | 0.2698<br>(0.4884)  |                        |
| HHI Sales Sup.-Ind.                                        |                                     |                       |                        |                     |                     |                     | -0.0846***<br>(-15.04) |
| Sup.-Ind.-Year FE                                          | Yes                                 | Yes                   | Yes                    | Yes                 | Yes                 | Yes                 | Yes                    |
| Cus.-Ind.-Year FE                                          | Yes                                 | Yes                   | Yes                    | Yes                 | Yes                 | Yes                 | Yes                    |
| Sup-Ctry-Cus-Ctry-Year FE                                  | Yes                                 | Yes                   | Yes                    | Yes                 | Yes                 | Yes                 | Yes                    |

Table 6: relationship and industry heterogeneity

## 5.6 Tables 7 and 8: climate adaptation and projections

### Read:

- Table 7a shows that the termination response is weaker in supplier countries with stronger ND-GAIN climate adaptation readiness.
- The economic readiness component has a particularly large negative interaction with exceeded expectations.
- Table 7b shows that customer performance effects of supplier heat remain even when separating heat days with and without country-level disaster declarations.
- Table 8 shows positive termination responses across suppliers located in areas with limited projected heat change under RCP2.6, RCP4.5, and RCP8.5.

**Economic message.** Country adaptation capacity dampens the supply-chain response, while long-horizon projections do not fully replace experienced climate signals.

Table 7

## Climate change adaptation in the supplier country

(a) Supplier country adaption readiness (ND-GAIN)

|                                                                                   | Dep. Var: $\mathbb{I}(\text{Last Relationship Year})$ |                      |                    |                    |
|-----------------------------------------------------------------------------------|-------------------------------------------------------|----------------------|--------------------|--------------------|
|                                                                                   | (1)                                                   | (2)                  | (3)                | (4)                |
| $\mathbb{I}(\text{Real.} > \text{Exp. Heat Days}) (> 1)$                          | 24.33***<br>(3.166)                                   | 29.19***<br>(3.072)  | 18.10**<br>(2.824) | 16.95**<br>(2.903) |
| $\mathbb{I}(\text{Real.} > \text{Exp. Heat Days}) (> 1) \times \text{Overall}$    | -23.25**<br>(-2.183)                                  |                      |                    |                    |
| $\mathbb{I}(\text{Real.} > \text{Exp. Heat Days}) (> 1) \times \text{Economic}$   |                                                       | -28.81**<br>(-2.377) |                    |                    |
| $\mathbb{I}(\text{Real.} > \text{Exp. Heat Days}) (> 1) \times \text{Governance}$ |                                                       |                      | -13.24<br>(-1.528) |                    |
| $\mathbb{I}(\text{Real.} > \text{Exp. Heat Days}) (> 1) \times \text{Social}$     |                                                       |                      |                    | -14.17<br>(-1.470) |
| Sup.-Ind.-Year FE                                                                 | Yes                                                   | Yes                  | Yes                | Yes                |
| Cus.-Ind.-Year FE                                                                 | Yes                                                   | Yes                  | Yes                | Yes                |
| Sup-Ctry-Cus-Ctry-Year FE                                                         | Yes                                                   | Yes                  | Yes                | Yes                |
| Observations                                                                      | 109,242                                               | 109,228              | 109,208            | 109,242            |
| $R^2$                                                                             | 0.1964                                                | 0.1966               | 0.1961             | 0.1963             |

(b) Country-level disaster declarations

|                                  | Cus Opt (t)           |                       |
|----------------------------------|-----------------------|-----------------------|
|                                  | (1)                   | (2)                   |
| Heat Days (t,t-3)                | -0.0002***<br>(-4.30) |                       |
| Heat Days No Declaration (t,t-3) |                       | -0.0002***<br>(-4.24) |
| Heat Days Declaration (t,t-3)    |                       | -0.0003***<br>(-3.90) |
| Firm $\times$ Fiscal-Qtr FE      | Yes                   | Yes                   |
| Ind $\times$ Year-Qtr FE         | Yes                   | Yes                   |
| Ctry-Linear-Trends               | Yes                   | Yes                   |
| BS2016 FE                        | Yes                   | Yes                   |
| Observations                     | 123700                | 123700                |
| Customers                        | 6299                  | 6299                  |

Table 7: adaptation readiness and disaster declarations

Table 8

## Experienced Exposure and Projections

|                                                          | Dep. Var.: $\mathbb{I}(\text{Last Relationship Year})$ |                           |                           |                     |
|----------------------------------------------------------|--------------------------------------------------------|---------------------------|---------------------------|---------------------|
|                                                          | $\approx 0$ RCP2.6<br>(1)                              | $\approx 0$ RCP4.5<br>(2) | $\approx 0$ RCP8.5<br>(3) | Full Sample<br>(4)  |
| $\mathbb{I}(\text{Real.} > \text{Exp. Heat Days}) (> 1)$ | 7.595***<br>(3.390)                                    | 8.474**<br>(3.008)        | 10.33**<br>(3.032)        | 7.853***<br>(5.595) |
| Sup.-Ind.-Year FE                                        | Yes                                                    | Yes                       | Yes                       | Yes                 |
| Cus.-Ind.-Year FE                                        | Yes                                                    | Yes                       | Yes                       | Yes                 |
| Sup-Ctry-Cus-Ctry-Year FE                                | Yes                                                    | Yes                       | Yes                       | Yes                 |
| Observations                                             | 45,324                                                 | 30,362                    | 18,563                    | 92,792              |
| $R^2$                                                    | 0.1715                                                 | 0.2183                    | 0.2563                    | 0.1488              |

This table presents linear probability model estimates on the effect of heat exceedance,  $\mathbb{I}(\text{Realized} > \text{Expected}) (> 1)$ , on the likelihood of supply-chain relationship termination. Estimates are presented separately for the full sample (column 4), as well as subsets of suppliers located in areas which are projected to experience

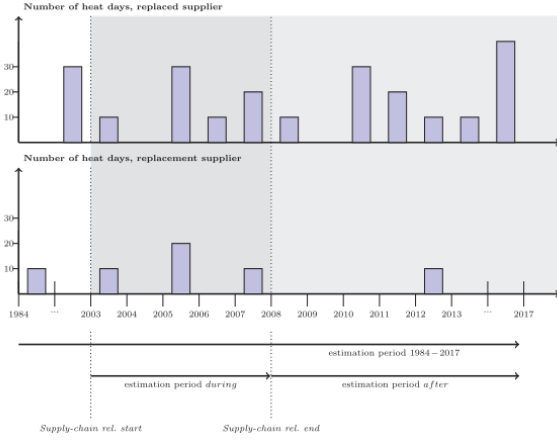
Table 8: experienced exposure and projections

## 5.7 Figure 4, Table 9, and Figure 5: supplier substitution

## Read:

- Figure 4 explains how the paper compares terminated suppliers with replacement suppliers during, after, and in projected future periods.
- Table 9 shows that after exceedance, replacement suppliers are more likely to have lower heat exposure than terminated suppliers during the initial relationship.
- The effect remains positive after the initial relationship, with coefficients around 0.095 and 0.056 across fixed-effect specifications.
- The projected-exposure result is positive in one specification but not robust in the specification with more fixed effects.
- Figure 5 shows the distributional version: replacement suppliers tend to be less exposed in realized historical periods, while projected exposure differences are weaker.

**Economic message.** The adaptation response affects not only exits but also the formation of new supply-chain links. Firms substitute toward suppliers that look less heat-exposed based on experienced data.



**Figure 4**  
**Variable construction: Exposure of replaced and replacement suppliers**  
This figure illustrates the construction of the exposure to heat of replaced and replacement suppliers (Table 9) based on a hypothetical example of a supplier and the replacement. We compare the heat days of old and new suppliers based on three time periods. First, we estimate and compare the exposure of the replaced and replacement supplier based on the years (*in dark gray*) during which the initial supply-chain relationship

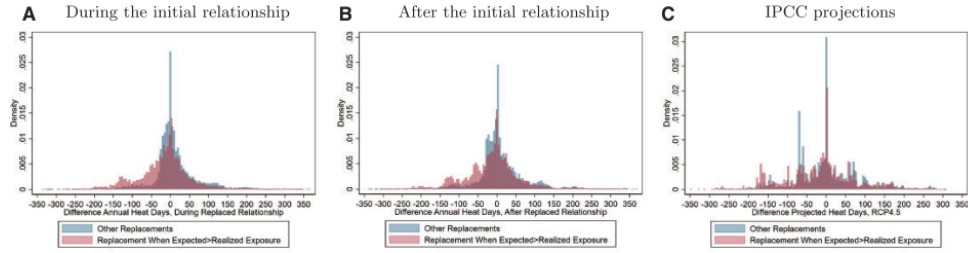
Figure 4: replacement-supplier comparison

**Table 9**  
**Heat exposure and supplier substitution**

|                             | 1(Decrease During Ini. Rel.) |                      | 1(Decrease After Ini. Rel.) |                      | 1(Decrease Projected Days) |                     |
|-----------------------------|------------------------------|----------------------|-----------------------------|----------------------|----------------------------|---------------------|
|                             | (1)                          | (2)                  | (3)                         | (4)                  | (5)                        | (6)                 |
| 1(Real.>Exp. Heat Days)     | 0.1508***<br>(7.535)         | 0.1229***<br>(7.635) | 0.0947***<br>(4.562)        | 0.0560***<br>(2.702) | 0.0543***<br>(2.864)       | -0.0188<br>(-1.111) |
| Cus.-Ind.-Year FE           | Yes                          | Yes                  | Yes                         | Yes                  | Yes                        | Yes                 |
| Sup.-Ind.-Year FE           | Yes                          | Yes                  | Yes                         | Yes                  | Yes                        | Yes                 |
| Sup.-Ctry-Cus.-Ctry-Year FE | No                           | Yes                  | No                          | Yes                  | No                         | Yes                 |
| Observations                | 17,178                       | 17,178               | 17,178                      | 17,178               | 17,178                     | 17,178              |
| R <sup>2</sup>              | 0.0771                       | 0.2475               | 0.0672                      | 0.2394               | 0.0560                     | 0.2429              |

This table shows the effect of the exceedance of the expected physical exposure to heat on supplier substitution. We match supplier firms for which the supplier-customer relationship is terminated during the sample period (*i.e.*, *old suppliers*) with their likely replacements (*i.e.*, *new suppliers*). Replacements are identified as firms with identical four-digit SIC codes, which enter a new supply-chain relationship with the given customer within one year of the termination of the previous supply-chain relationship. The results show linear probability model estimates on the likelihood that the average to heat days of new replacement suppliers is lower than the average

Table 9: heat exposure and substitution



**Figure 5**  
**Heat exposure and supplier substitution**  
This figure shows the effect of an exceedance of ex ante expectations of heat exposure during the initial relationship on the exposure of new suppliers, analogous to Table 9. It plots the distribution of the difference in annual heat days between the original supplier and the potential replacement, identified based on four-digit SIC codes and a start date in the year in which the initial relationship is terminated. The figures separately plot the distribution for supplier replacement cases in which realized heat days exceeded expected heat days (red), as these terminations could have been related to climate-change-related considerations. The blue distribution plots all other supplier replacements. In terms of the affected time periods, the figures compare the exposure of initial and replacement suppliers during (Figure A) and after (Figure B) the initial supply-chain relationship, and compares projected exposures according to the MIP5 RCP4.5 projections (Figure C).

## 6 Short summary

- The paper studies climate-change adaptation in global supply chains.
- Supplier heat exposure reduces supplier operating income and propagates to customers.
- The main adaptation measure compares realized heat exposure during a relationship with historical expected exposure before the relationship.
- Customers are more likely to terminate supplier relationships when realized exposure exceeds expectations.
- The effect is stronger when the signal is repeated or larger, consistent with experience-based learning.
- Transitory heat days alone do not explain terminations.
- Customers also increase inventories, COGS, R&D, cash, and supplier diversification.
- Termination is more likely when switching costs are lower and substitutes are easier to find.
- Adaptation readiness in supplier countries weakens the termination response.

- Customers replace exposed suppliers with new suppliers that have lower expected and realized heat exposure.
- Long-run climate projections are less central to the substitution result than experienced heat realizations.
- The results suggest that climate change reshapes production networks through both performance shocks and endogenous firm adaptation.

## 7 Conclusion

### 7.1 Core conclusion

Pankratz and Schiller show that physical climate risk affects firms through global supply chains. Heat exposure at suppliers reduces supplier performance, propagates to customers, and triggers customer adaptation when realized exposure exceeds historical expectations.

### 7.2 Economic intuition

Customers form expectations about supplier climate exposure when relationships begin. When experienced heat exceeds those priors, the supplier may no longer be optimal. Because supply-chain relationships are costly to maintain and disruptions can harm performance, customers update beliefs and adjust by terminating suppliers, diversifying, holding buffers, and choosing less exposed replacements.

### 7.3 Takeaway

The paper connects climate finance with production networks. Climate change is not only a local productivity shock: it changes how firms learn, contract, and choose suppliers across borders. The main lesson is that adaptation occurs through network reallocation, not only through local investment or insurance.